Newman-Penrose Formalism Calculations

Stephani et al., Equation (16.19)

Metric

$$
\begin{aligned}
ds^2 = &\left(-\frac{\omega_p \sin\left((R-R_m)\omega_m\right) - \omega_m \sin\left((R-R_p)\omega_p\right)}{\omega_p \sin\left((R-R_m)\omega_m\right) + \omega_m \sin\left((R-R_p)\omega_p\right)}\right) \mathrm{d}t \otimes \mathrm{d}t + \left(\frac{\omega_p \sin\left((R-R_m)\omega_m\right) + \omega_m \sin\left((R-R_p)\omega_p\right)}{\omega_p \sin\left((R-R_m)\omega_m\right) - \omega_m \sin\left((R-R_p)\omega_p\right)}\right) \mathrm{d}R \otimes \mathrm{d}R \\
&+ \left(\frac{\omega_p^2 \cosh(c)^2 \sin\left((R-R_m)\omega_m\right)^2 + 2\,\omega_m\omega_p \cosh(c)^2 \sin\left((R-R_m)\omega_m\right) \sin\left((R-R_p)\omega_p\right) + \omega_m^2 \cosh(c)^2 \sin\left((R-R_p)\omega_p\right)^2}{\omega_m^2\omega_p^2}\right) \mathrm{d}\theta \otimes \mathrm{d}\theta \\
&+ \frac{\left(\omega_p^2 \cosh(c)^2 \sin\left((R-R_m)\omega_m\right)^2 + 2\,\omega_m\omega_p \cosh(c)^2 \sin\left((R-R_m)\omega_m\right) \sin\left((R-R_p)\omega_p\right) + \omega_m^2 \cosh(c)^2 \sin\left((R-R_p)\omega_p\right)^2\right) \sin(\theta)^2}{\omega_m^2\omega_p^2} \mathrm{d}\phi \otimes \mathrm{d}\phi
\end{aligned}
$$

Coordinates

$(x^\mu) = (t, R, \theta, \phi)$

Metric Functions

$$A = \frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m}$$

$$B = \frac{\cosh(c) \sin(R\omega_p - R_p\omega_p)}{\omega_p}$$

$$Z = \frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m\left(\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}\right)} - \frac{\cosh(c) \sin(R\omega_p - R_p\omega_p)}{\omega_p\left(\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}\right)}$$

$$V = \frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} - \frac{\cosh(c) \sin(R\omega_p - R_p\omega_p)}{\omega_p}$$

$$W = \frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}$$

Null Tetrad

Covariant components

$$l = \left(-\sqrt{\frac{1}{2}}\sqrt{\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m\left(\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}\right)} - \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p\left(\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}\right)}}\right) dt$$
$$+ \left(\frac{\sqrt{2}}{2\sqrt{\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m\left(\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}\right)} - \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p\left(\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}\right)}}}\right) dR$$

$$n = \left(-\sqrt{\frac{1}{2}}\sqrt{\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m\left(\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}\right)} - \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p\left(\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}\right)}}\right) dt$$
$$+ \left(-\frac{\sqrt{2}}{2\sqrt{\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m\left(\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}\right)} - \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p\left(\frac{\cosh(c)\sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\cosh(c)\sin(R\omega_p - R_p\omega_p)}{\omega_p}\right)}}}\right) dR$$

$$m = \left(\frac{\sqrt{2}\cosh(c)\sin(R\omega_m - R_m\omega_m)}{2\,\omega_m} + \frac{\sqrt{2}\cosh(c)\sin(R\omega_p - R_p\omega_p)}{2\,\omega_p}\right) d\theta + \left(\frac{i\sqrt{2}\cosh(c)\sin(R\omega_m - R_m\omega_m)\sin(\theta)}{2\,\omega_m} + \frac{i\sqrt{2}\cosh(c)\sin(R\omega_p - R_p\omega_p)\sin(\theta)}{2\,\omega_p}\right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}\cosh(c)\sin(R\omega_m - R_m\omega_m)}{2\,\omega_m} + \frac{\sqrt{2}\cosh(c)\sin(R\omega_p - R_p\omega_p)}{2\,\omega_p}\right) d\theta + \left(-\frac{i\sqrt{2}\cosh(c)\sin(R\omega_m - R_m\omega_m)\sin(\theta)}{2\,\omega_m} - \frac{i\sqrt{2}\cosh(c)\sin(R\omega_p - R_p\omega_p)\sin(\theta)}{2\,\omega_p}\right) d\phi$$

Contravariant components

$$l = \left(\frac{\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}}{\sqrt{2}\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \sqrt{2}\omega_m \sin\left(R\omega_p - R_p\omega_p\right)} \right) dt$$
$$+ \left(\frac{\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}}{\sqrt{2}\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \sqrt{2}\omega_m \sin\left(R\omega_p - R_p\omega_p\right)} \right) dR$$

$$n = \left(\frac{\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}}{\sqrt{2}\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \sqrt{2}\omega_m \sin\left(R\omega_p - R_p\omega_p\right)} \right) dt$$
$$+ \left(-\frac{\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}}{\sqrt{2}\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \sqrt{2}\omega_m \sin\left(R\omega_p - R_p\omega_p\right)} \right) dR$$

$$m = \left(\frac{\omega_m\omega_p}{\sqrt{2}\omega_p \cosh\left(c\right) \sin\left(R\omega_m - R_m\omega_m\right) + \sqrt{2}\omega_m \cosh\left(c\right) \sin\left(R\omega_p - R_p\omega_p\right)} \right) d\theta + \left(\frac{i\,\omega_m\omega_p}{\left(\sqrt{2}\omega_p \cosh\left(c\right) \sin\left(R\omega_m - R_m\omega_m\right) + \sqrt{2}\omega_m \cosh\left(c\right) \sin\left(R\omega_p - R_p\omega_p\right)\right) \sin\left(\theta\right)} \right) d\phi$$

$$\overline{m} = \left(\frac{\omega_m\omega_p}{\sqrt{2}\omega_p \cosh\left(c\right) \sin\left(R\omega_m - R_m\omega_m\right) + \sqrt{2}\omega_m \cosh\left(c\right) \sin\left(R\omega_p - R_p\omega_p\right)} \right) d\theta + \left(-\frac{i\,\omega_m\omega_p}{\left(\sqrt{2}\omega_p \cosh\left(c\right) \sin\left(R\omega_m - R_m\omega_m\right) + \sqrt{2}\omega_m \cosh\left(c\right) \sin\left(R\omega_p - R_p\omega_p\right)\right) \sin\left(\theta\right)} \right) d\phi$$

Directional Derivatives

$$D = \frac{\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}}{\sqrt{2}\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \sqrt{2}\omega_m \sin\left(R\omega_p - R_p\omega_p\right)} \partial_t$$
$$+ \frac{\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}}{\sqrt{2}\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \sqrt{2}\omega_m \sin\left(R\omega_p - R_p\omega_p\right)} \partial_R$$

$$\Delta = \frac{\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}}{\sqrt{2}\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \sqrt{2}\omega_m \sin\left(R\omega_p - R_p\omega_p\right)} \partial_t$$
$$- \frac{\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}\sqrt{\omega_p \sin\left(R\omega_m - R_m\omega_m\right) - \omega_m \sin\left(R\omega_p - R_p\omega_p\right)}}{\sqrt{2}\omega_p \sin\left(R\omega_m - R_m\omega_m\right) + \sqrt{2}\omega_m \sin\left(R\omega_p - R_p\omega_p\right)} \partial_R$$

$$\delta = \frac{\omega_m\omega_p}{\sqrt{2}\omega_p \cosh\left(c\right) \sin\left(R\omega_m - R_m\omega_m\right) + \sqrt{2}\omega_m \cosh\left(c\right) \sin\left(R\omega_p - R_p\omega_p\right)} \partial_\theta + \frac{i\,\omega_m\omega_p}{\left(\sqrt{2}\omega_p \cosh\left(c\right) \sin\left(R\omega_m - R_m\omega_m\right) + \sqrt{2}\omega_m \cosh\left(c\right) \sin\left(R\omega_p - R_p\omega_p\right)\right) \sin\left(\theta\right)} \partial_\phi$$

$$\overline{\delta} = \frac{\omega_m \omega_p}{\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \sqrt{2}\omega_m \cosh(c) \sin(R\omega_p - R_p\omega_p)} \partial_\theta - \frac{i\,\omega_m \omega_p}{\left(\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \sqrt{2}\omega_m \cosh(c) \sin(R\omega_p - R_p\omega_p)\right) \sin(\theta)} \partial_\phi$$

Spin Coefficients

$$\begin{aligned}
\rho = &-\frac{\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)}\omega_m \omega_p^2 \cos(R\omega_m - R_m\omega_m) \sin(R\omega_m - R_m\omega_m)}{\left(\sqrt{2}\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\sqrt{2}\omega_m\omega_p \sin(R\omega_m - R_m\omega_m) \sin(R\omega_p - R_p\omega_p) + \sqrt{2}\omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}} \\
&- \frac{\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)}\omega_m \omega_p^2 \cos(R\omega_p - R_p\omega_p) \sin(R\omega_m - R_m\omega_m)}{\left(\sqrt{2}\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\sqrt{2}\omega_m\omega_p \sin(R\omega_m - R_m\omega_m) \sin(R\omega_p - R_p\omega_p) + \sqrt{2}\omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}} \\
&+ \frac{\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)}\omega_m^2 \omega_p \cos(R\omega_m - R_m\omega_m) \sin(R\omega_p - R_p\omega_p)}{\left(\sqrt{2}\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\sqrt{2}\omega_m\omega_p \sin(R\omega_m - R_m\omega_m) \sin(R\omega_p - R_p\omega_p) + \sqrt{2}\omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}} \\
&+ \frac{\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)}\omega_m^2 \omega_p \cos(R\omega_p - R_p\omega_p) \sin(R\omega_p - R_p\omega_p)}{\left(\sqrt{2}\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\sqrt{2}\omega_m\omega_p \sin(R\omega_m - R_m\omega_m) \sin(R\omega_p - R_p\omega_p) + \sqrt{2}\omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}}
\end{aligned}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\begin{aligned}
\mu = &-\frac{\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)}\omega_m \omega_p^2 \cos(R\omega_m - R_m\omega_m) \sin(R\omega_m - R_m\omega_m)}{\left(\sqrt{2}\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\sqrt{2}\omega_m\omega_p \sin(R\omega_m - R_m\omega_m) \sin(R\omega_p - R_p\omega_p) + \sqrt{2}\omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}} \\
&- \frac{\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)}\omega_m \omega_p^2 \cos(R\omega_p - R_p\omega_p) \sin(R\omega_m - R_m\omega_m)}{\left(\sqrt{2}\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\sqrt{2}\omega_m\omega_p \sin(R\omega_m - R_m\omega_m) \sin(R\omega_p - R_p\omega_p) + \sqrt{2}\omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}} \\
&+ \frac{\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)}\omega_m^2 \omega_p \cos(R\omega_m - R_m\omega_m) \sin(R\omega_p - R_p\omega_p)}{\left(\sqrt{2}\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\sqrt{2}\omega_m\omega_p \sin(R\omega_m - R_m\omega_m) \sin(R\omega_p - R_p\omega_p) + \sqrt{2}\omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}} \\
&+ \frac{\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)}\omega_m^2 \omega_p \cos(R\omega_p - R_p\omega_p) \sin(R\omega_p - R_p\omega_p)}{\left(\sqrt{2}\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\sqrt{2}\omega_m\omega_p \sin(R\omega_m - R_m\omega_m) \sin(R\omega_p - R_p\omega_p) + \sqrt{2}\omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}}
\end{aligned}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\omega_m \omega_p \cos(\theta)}{2\left(\sqrt{2}\omega_p \cosh(c)\sin(R\omega_m - R_m\omega_m) + \sqrt{2}\omega_m \cosh(c)\sin(R\omega_p - R_p\omega_p)\right)\sin(\theta)}$$

$$\beta = \frac{\omega_m \omega_p \cos(\theta)}{2\left(\sqrt{2}\omega_p \cosh(c)\sin(R\omega_m - R_m\omega_m) + \sqrt{2}\omega_m \cosh(c)\sin(R\omega_p - R_p\omega_p)\right)\sin(\theta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\begin{aligned}\epsilon = &-\frac{\sqrt{2}\omega_m \omega_p^2 \cos(R\omega_p - R_p\omega_p)\sin(R\omega_m - R_m\omega_m)}{4\left(\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)\right)^{\frac{3}{2}}\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}} \\ &+ \frac{\sqrt{2}\omega_m^2 \omega_p \cos(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)}{4\left(\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)\right)^{\frac{3}{2}}\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}}\end{aligned}$$

$$\begin{aligned}\gamma = &-\frac{\sqrt{2}\omega_m \omega_p^2 \cos(R\omega_p - R_p\omega_p)\sin(R\omega_m - R_m\omega_m)}{4\left(\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)\right)^{\frac{3}{2}}\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}} \\ &+ \frac{\sqrt{2}\omega_m^2 \omega_p \cos(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)}{4\left(\omega_p \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p)\right)^{\frac{3}{2}}\sqrt{\omega_p \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p)}}\end{aligned}$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = &-\frac{\omega_m^2 \omega_p^2 \cos(R\omega_m - R_m\omega_m)^2}{2\left(\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\,\omega_m\omega_p \sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p) + \omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)} \\ &+ \frac{\omega_m^2 \omega_p^2 \cos(R\omega_p - R_p\omega_p)^2}{2\left(\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\,\omega_m\omega_p \sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p) + \omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)} \\ &- \frac{\omega_m^3 \omega_p \sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)}{2\left(\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\,\omega_m\omega_p \sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p) + \omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)} \\ &+ \frac{\omega_m \omega_p^3 \sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)}{2\left(\omega_p^2 \sin(R\omega_m - R_m\omega_m)^2 + 2\,\omega_m\omega_p \sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p) + \omega_m^2 \sin(R\omega_p - R_p\omega_p)^2\right)}\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}
\Phi_{11} = &-\frac{\omega_m^2\omega_p^2\cos\left(R\omega_m - R_m\omega_m\right)^2\cosh\left(c\right)^2}{4\left(\omega_p^2\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\cosh\left(c\right)^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)} \\
&+ \frac{\omega_m^2\omega_p^2\cos\left(R\omega_p - R_p\omega_p\right)^2\cosh\left(c\right)^2}{4\left(\omega_p^2\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\cosh\left(c\right)^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)} \\
&- \frac{\omega_m^3\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right)}{4\left(\omega_p^2\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\cosh\left(c\right)^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)} \\
&+ \frac{\omega_m\omega_p^3\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right)}{4\left(\omega_p^2\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\cosh\left(c\right)^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)} \\
&+ \frac{\omega_m^2\omega_p^2}{4\left(\omega_p^2\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\cosh\left(c\right)^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)}
\end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{\omega_m^2\omega_p^2\cos\left(R\omega_m - R_m\omega_m\right)^2}{2\left(\omega_p^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)}$$
$$+\frac{\omega_m^2\omega_p^2\cos\left(R\omega_p - R_p\omega_p\right)^2}{2\left(\omega_p^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)}$$
$$-\frac{\omega_m^3\omega_p\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right)}{2\left(\omega_p^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)}$$
$$+\frac{\omega_m\omega_p^3\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right)}{2\left(\omega_p^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)}$$

$$\Lambda = -\frac{\omega_m^2\omega_p^2\cos\left(R\omega_m - R_m\omega_m\right)^2\cosh\left(c\right)^2}{4\left(\omega_p^2\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\cosh\left(c\right)^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)}$$
$$+\frac{\omega_m^2\omega_p^2\cos\left(R\omega_p - R_p\omega_p\right)^2\cosh\left(c\right)^2}{4\left(\omega_p^2\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\cosh\left(c\right)^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)}$$
$$-\frac{\omega_m^3\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right)}{12\left(\omega_p^2\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\cosh\left(c\right)^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)}$$
$$+\frac{\omega_m\omega_p^3\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right)}{12\left(\omega_p^2\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\cosh\left(c\right)^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)}$$
$$+\frac{\omega_m^2\omega_p^2}{12\left(\omega_p^2\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)^2 + 2\,\omega_m\omega_p\cosh\left(c\right)^2\sin\left(R\omega_m - R_m\omega_m\right)\sin\left(R\omega_p - R_p\omega_p\right) + \omega_m^2\cosh\left(c\right)^2\sin\left(R\omega_p - R_p\omega_p\right)^2\right)}$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$
\begin{aligned}
\Psi_2 = & \frac{\omega_m^2\,\omega_p^3\cos(R\omega_m - R_m\omega_m)\cos(R\omega_p - R_p\omega_p)\cosh(c)^2\sin(R\omega_m - R_m\omega_m)}{\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^3 + 3\,\omega_m\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p) + 3\,\omega_m^2\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2 + \omega_m^3\cosh(c)^2\sin(R\omega_p - R_p\omega_p)^3} \\
& - \frac{\omega_m^3\omega_p^2\cos(R\omega_m - R_m\omega_m)\cos(R\omega_p - R_p\omega_p)\cosh(c)^2\sin(R\omega_p - R_p\omega_p)}{\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^3 + 3\,\omega_m\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p) + 3\,\omega_m^2\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2 + \omega_m^3\cosh(c)^2\sin(R\omega_p - R_p\omega_p)^3} \\
& + \frac{2\,\omega_m^3\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p)}{3\,(\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^3 + 3\,\omega_m\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p) + 3\,\omega_m^2\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2 + \omega_m^3\cosh(c)^2\sin(R\omega_p - R_p\omega_p)^3)} \\
& + \frac{\omega_m\omega_p^4\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p)}{3\,(\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^3 + 3\,\omega_m\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p) + 3\,\omega_m^2\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2 + \omega_m^3\cosh(c)^2\sin(R\omega_p - R_p\omega_p)^3)} \\
& - \frac{\omega_m^4\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2}{3\,(\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^3 + 3\,\omega_m\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p) + 3\,\omega_m^2\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2 + \omega_m^3\cosh(c)^2\sin(R\omega_p - R_p\omega_p)^3)} \\
& - \frac{2\,\omega_m^2\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2}{3\,(\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^3 + 3\,\omega_m\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p) + 3\,\omega_m^2\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2 + \omega_m^3\cosh(c)^2\sin(R\omega_p - R_p\omega_p)^3)} \\
& + \frac{\omega_m^2\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)}{\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^3 + 3\,\omega_m\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p) + 3\,\omega_m^2\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2 + \omega_m^3\cosh(c)^2\sin(R\omega_p - R_p\omega_p)^3} \\
& - \frac{\omega_m^3\omega_p^2\cosh(c)^2\sin(R\omega_p - R_p\omega_p)}{\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^3 + 3\,\omega_m\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p) + 3\,\omega_m^2\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2 + \omega_m^3\cosh(c)^2\sin(R\omega_p - R_p\omega_p)^3} \\
& - \frac{\omega_m^2\omega_p^3\sin(R\omega_m - R_m\omega_m)}{6\,(\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^3 + 3\,\omega_m\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p) + 3\,\omega_m^2\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2 + \omega_m^3\cosh(c)^2\sin(R\omega_p - R_p\omega_p)^3)} \\
& - \frac{\omega_m^3\omega_p^2\sin(R\omega_p - R_p\omega_p)}{6\,(\omega_p^3\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^3 + 3\,\omega_m\omega_p^2\cosh(c)^2\sin(R\omega_m - R_m\omega_m)^2\sin(R\omega_p - R_p\omega_p) + 3\,\omega_m^2\omega_p\cosh(c)^2\sin(R\omega_m - R_m\omega_m)\sin(R\omega_p - R_p\omega_p)^2 + \omega_m^3\cosh(c)^2\sin(R\omega_p - R_p\omega_p)^3)}
\end{aligned}
$$

$\Psi_3 = 0$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"